

A Project Report on
Data Visualization With Power BI



FACULTY OF COMPUTER APPLICATIONS

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my teacher for giving me the opportunity to create this mini project on topic “Mental Health Analysis using Power BI”. This project helped me understand how data visualization and analysis are used to find meaningful insights from data.

I also learned how to create dashboards, charts, slicers, and reports in Power BI. This project improved my analytical thinking and practical skills.

INTRODUCTION

Mental health has become an important topic in today's world. Stress, anxiety, depression, sleep patterns, and work pressure affect people's daily lives and productivity. Organization and researchers use data analysis to understand these problems and make better decisions.

This project focuses on analyzing mental health data using Power BI. Different visualizations are used to study stress levels, anxiety scores, depression scores, sleep hours, work environments, and treatment behavior . The dashboard helps in understanding patterns and relationships between different factors.

Power BI makes the data easy to understand through interactive data and visuals.

OBJECTIVES OF THE PROJECT

The main objective of this project are:

To analyze mental health data using Power BI

- To study stress, anxiety, and depression levels
- To understand the effect of sleep on stress and productivity
- To compare productivity in different work environments
- To analyze gender-wise mental health conditions
- To create interactive dashboards for better visualization

TOOLS AND TECHNOLOGIES USED

Tool

Purpose

Power BI

Dashboard creation
And Visualization

Excel/CSV Dataset

Data source

Data Analysis

Finding insights from
data

	gender	employment_status	work_environment	mental_health_hist	seeks_treatment	stress_level	sleep_hours	physical_activity_d	depression_score	anxiety_score	social_support_score	productivity_score	mental_health_risk
	56 Male	Employed	On-site	Yes	Yes	6	6.2	3	28	17	54	59.7	High
	46 Female	Student	On-site	No	Yes	10	9	4	30	11	85	54.9	High
	32 Female	Employed	On-site	Yes	No	7	7.7	2	24	7	62	61.3	Medium
	60 Non-binary	Self-employed	On-site	No	No	4	4.5	4	6	0	95	97	Low
	25 Female	Self-employed	On-site	Yes	Yes	3	5.4	0	24	12	70	69	High
	38 Female	Unemployed	On-site	Yes	Yes	3	9.9	3	17	9	63	69.3	Medium
	56 Non-binary	Employed	Remote	No	Yes	2	5.5	1	25	3	87	63.1	Medium
	36 Male	Student	Remote	No	No	8	7.1	5	25	8	72	58.4	Medium
	40 Male	Employed	Hybrid	No	No	7	6.5	6	28	6	46	60.4	Medium
	28 Female	Employed	Remote	No	No	8	3.1	0	7	18	88	93.2	Medium
	28 Female	Self-employed	On-site	Yes	No	6	9.7	2	14	9	85	71.2	Medium
	41 Female	Self-employed	On-site	Yes	No	10	4.4	7	13	2	23	80.1	Medium
	53 Male	Self-employed	Remote	No	Yes	10	6.5	1	15	17	43	79.6	Medium
	57 Male	Employed	On-site	No	No	5	5.1	6	5	7	6	93	Low
	41 Male	Employed	On-site	No	No	1	8	1	22	0	52	75.9	Medium
	20 Male	Employed	On-site	No	Yes	8	6.7	3	22	17	22	59.1	High
	39 Male	Employed	Remote	No	No	2	8.7	2	14	6	64	85.8	Medium
	19 Female	Employed	Hybrid	No	Yes	1	9.8	5	21	21	36	64.1	High
	41 Male	Employed	Hybrid	No	Yes	9	8.2	5	3	14	99	84.8	Medium
	61 Female	Unemployed	On-site	No	No	8	7.3	6	2	8	68	98.6	Low
	47 Male	Student	On-site	Yes	No	9	5.6	2	12	8	22	81.4	Medium
	55 Female	Student	Hybrid	Yes	No	5	5.8	1	11	4	41	84	Medium
	19 Male	Student	Remote	Yes	No	3	7.8	2	8	12	84	90.6	Medium
	38 Male	Employed	On-site	No	No	8	6.5	7	4	21	14	96.7	Medium
	50 Male	Employed	Remote	No	No	4	8.9	4	11	12	65	80.8	Medium
	29 Prefer not to say	Student	Remote	No	No	1	8.9	3	28	21	84	62.1	High
	39 Female	Employed	Hybrid	No	No	9	7	7	0	2	17	89.9	Low
	61 Female	Employed	Remote	No	No	4	5.3	3	0	18	39	99.5	Medium
	42 Male	Employed	On-site	Yes	No	2	6.2	4	23	11	17	71.1	Medium

DATASET DESCRIPTION

The dataset contains information related to the mental health and productivity of individuals.

Columns Used in Dataset

Column Name	Description
age	Age of the person
gender	Gender category
stress_level	Stress Score
anxiety_score	Anxiety level
depression _score	Depression level
sleep_hours	Average sleep hours
productivity_score	Productivity score
work _environment	Remote, Hybrid, or On-site

DATA CLENNING PROCESS

Before creating the dashboard, the dataset was cleaned and prepared properly.

The following steps were performed:

- Checked missing values
- Verified data types
- Removed duplicate records
- Organized columns properly
- Corrected formatting issues
- Imported clean data into Power BI

Data cleaning improved the accuracy and quality of the dashboard.

"Mental Health Analysis Dashboard"

5.55

Average of stres...

15.01

Average of depression.

6.50

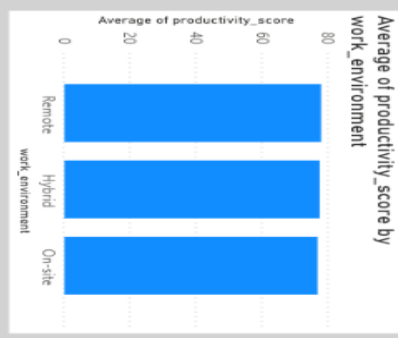
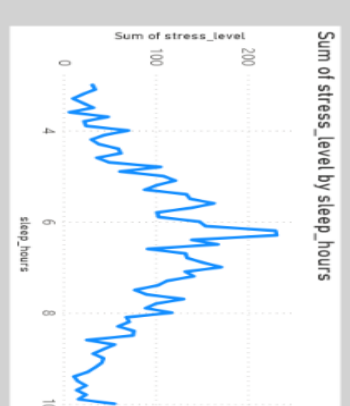
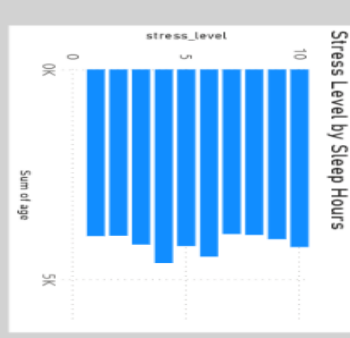
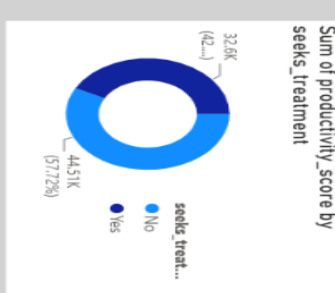
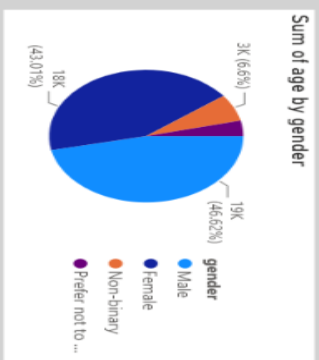
Average of sleep_hour

10.50

Average of anxiety_sco..

value

↓



OVERVIEW DASHBOARD

The Overview Dashboard provides a summary of the complete dataset through different visualizations.

Visuals Used

- Average Stress Level
- Average Depression Score
- Average Sleep Hours
- Average Anxiety Score
- Gender Distribution Chart
- Stress Level by Sleep Hours
- Productivity Score by Work Environment
- Treatment Distribution Chart

Dashboard Analysis

The dashboard shows that stress, anxiety, and depression are important mental health factors affecting individuals. The average sleep hours are around 6.5 hours, which directly impacts stress and productivity levels.

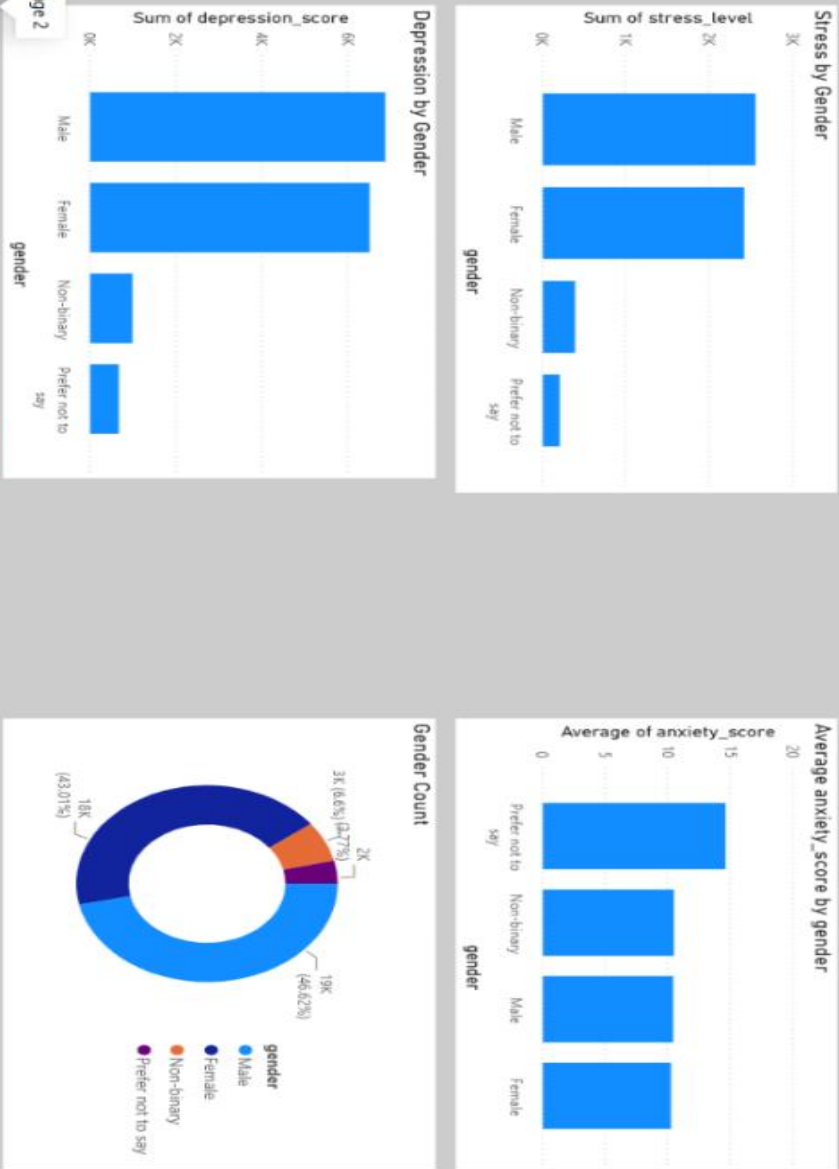
The dashboard also shows that productivity changes according to different work environments.

Interactive slicers help users filter data based on gender and work environment.

Insights

- Average stress level is moderate among individuals.
- Sleep patterns affect stress levels.
- Work environment impacts productivity.

GENDER ANALYSIS



GENDER ANALYSIS

This page focuses on analyzing mental health conditions according to gender categories.

Visuals Used

- ❖ Stress by Gender
- ❖ Anxiety Score by Gender
- ❖ Depression Score by Gender

Dashboard Analysis

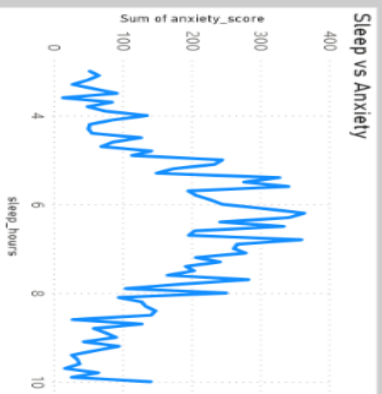
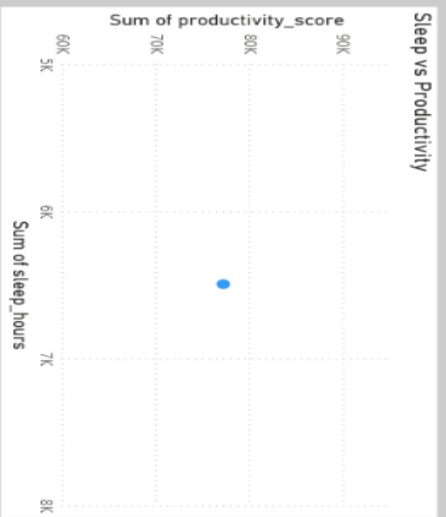
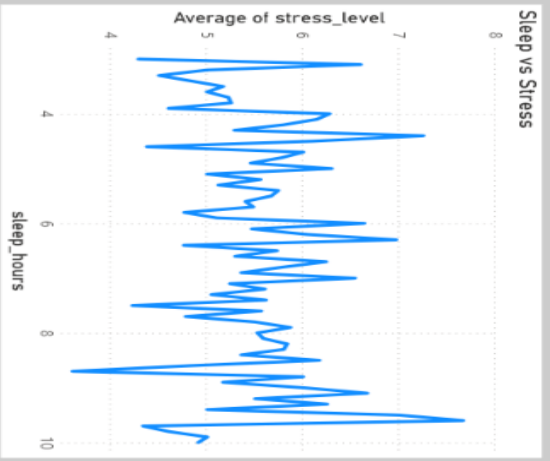
This dashboard compares stress, anxiety, and depression scores among different gender groups. It helps in understanding how mental health conditions vary among individuals.

The gender distribution chart shows participation from multiple gender categories.

Insights

- Male and female participants show different stress patterns.
- Anxiety and depression levels vary by gender.
- Some gender groups reported higher mental health concerns.

"SLEEP ANALYSIS"



Lower sleep hours
increase stress and
anxiety levels.

SLEEP ANALYSIS

This page analyzes the relationship between sleep and mental health.

Visuals Used

- ❖ Stress Level by sleep Hours
- ❖ Anxiety Score by Sleep Hours
- ❖ Productivity Score by Sleep now Hours

Dashboard Analysis

Sleep plays an important role in mental and physical well-being. This dashboard helps in identifying how sleep affects stress, anxiety, and productivity.

The charts show that lower sleep hours are connected with higher stress levels.

Insights

- Proper sleep improves productivity.
- Less sleep increases stress and anxiety.
- Balanced sleep patterns support better mental health.

FileHomeInsertModelingViewOptimizeHelp

Clipboard

CutCopyPasteFormat painter

Get dataExcelOnLake SQL Enter Datasource Recent workbook catalog Server data sources

Transform Refresh data

Queries

InsertNew Text box visualMore visuals

CalculationsNew visualNew measureQuick calculation

Sensitivity

SharePublish

CopilotPrep data for Copilot AI

Share

Stress by work environment

stress_level

Count of work_environment

Depression by work environment

Sum of depression_score

work_environment

"WORK ENVIRONMENT"

. Work pressure affects stress levels.
. Mental health conditions vary across different work environment.

Page 1Page 2Page 3Page 4Page 5Page 6

Page 4

Page 4 of 6

83%

WORK ENVIRONMENT ANALYSIS

The page focuses on comparing productivity's and stress in different work environments.

Visuals Used

- ❖ Productivity by Work Environment
- ❖ Stress Level by Work Environment
- ❖ Depression Score by Work Environment
- ❖ Anxiety Score by work Environment

Dashboard Analysis

Different work environment affect mental health and productivity in different ways. This dashboard compares remote, hybrid, and on-site work conditions. The analysis helps in understanding employee well-being and performance.

Insights

- ❖ Remote and hybrid work environments show good productivity.
- ❖ Work pressure affects stress levels.
- ❖ Mental health conditions vary across different work environments.



TREATMENT ANALYSIS

The page analyzes treatment – seeking behaviour among individuals.

Visuals used

- ❖ Treatment Distribution Chart
- ❖ Treatment by Gender
- ❖ Stress Level by Treatments status

Dashboard Analysis

This dashboard helps in understanding whether individuals seek treatment for mental health issues. It also compares stress levels among people who seek treatment and those who do not.

Insights

- ❖ People with higher stress levels are more likely to seek treatment.
- ❖ Mental health awareness is important.
- ❖ Treatment support can improve well-being and productivity -well-being and productivity.

CONCLUSION DASHBOARD

The final dashboard summarizes the important findings of the project.

Key Findings

- Sleep patterns strongly affect stress and productivity.
- Mental health conditions vary across genders and work environments.
- Stress, anxiety, and depression directly impact productivity.
- Data visualization makes complex information easy to understand.

The Conclusion Dashboard summarizes the overall finding and insights of the Mental Health Analysis project. This page combines the important observations from all previous dashboards and provides a better understanding of how stress, anxiety, depression, sleep patterns and work environments affect individuals.

The analysis shows that mental health conditions are closely connected with productivity and daily lifestyle. Individuals with proper sleep patterns generally experience lower stress levels and better productivity, while lack of sleep increases stress and anxiety.

The project also highlights that work environment influences employee performance and mental well-being. Different gender groups showed different stress and anxiety patterns, which indicates that mental health experience vary among individuals.

The treatment analysis revealed that work people with higher stress levels are more likely to seek mental health support and treatment. This shows the importance of awareness and proper care regarding mental health issues.

Power BI helped in converting raw data into meaningful visuals insights through interactive charts, KPI cards, slicers, and dashboards. The project improved understanding of data visualization and practical analysis techniques.

FEATURES OF THE DASHBOARD

The dashboards contains the following features:

- ✓ Interactive slicers
- ✓ Multiple charts and visualizations
- ✓ User-friendly layout
- ✓ Clear titles and labels
- ✓ Easy data filtering
- ✓ Professional dashboard design

CONCLUSION

The “Mental Health Analysis Dashboard” successfully represents important mental health data using Power BI visualizations. The project shows the relationship between stress, anxiety, depression, sleep patterns, productivity, and work environments.

Power BI helped in creating interactive dashboards that make data analysis easier and more understandable. This project provided practical experience in data visualization and analysis.

FUTURE SCOPE

This project can be improved further by:

- ✓ Using real-time datasets
- ✓ Adding machine learning predictions
- ✓ Creating advanced AI-based dashboards
- ✓ Including more health-related factors

REFERENCES

- Power BI datasets
- Kaggle Dataset
- Online Learning Resources

LEARNINGS FROM THE PROJECT

- This project helped me learn:
- Data visualization using Power BI
- Dashboard designing
- Data analysis techniques
- Importance of data cleaning
- Creating interactive reports

The project also improved my analytical thinking and presentation skills.

FINAL CONCLUSION

The mental Health Analysis Dashboard successfully represents important mental health data using Power BI visualizations. This project helped in understanding the relationship between stress, sleep, productivity, and work environments through data analysis and interactive dashboards.